1. Consider a duopoly market with inverse demand $p = a - (q_1 + q_2)$. The total cost for each firm is given by $C(q) = cq$. Suppose that each firm has only two quantity-choices: either $(a-c)/4$ or $(a-c)/3$. No other quantities are feasible. Write down the normal form of this game. Show that this two-action game is equivalent to the Prisoners' Dilemma: each firm has a strictly dominated strategy and both are worse off in equilibrium than they would be if they cooperated.

2. In a game between the government (player 1) and an individual (player 2), player 2 chooses an action a_2 which is either 0 or 1 and receives a transfer t which is also either 0 or 1 from the government. The sequence of the game is as follows: the government first announces one of the four possible pure transfer rules $t(a_2)$ (e.g., (1, 0) which implies the transfer is 1 if player 2's action is 0 and 0 if the action is 1), and then player 2 chooses one of his two possible actions, 0 and 1. The government is constrained to implement the transfer rule it announced. Player 2's objective is to maximize the value of his transfer, minus the cost of his action, which is 0 for $a_2 = 0$ and ½ for $a_2 = 1$. Player 1's objective is to minimize the sum $2(a_2 - 1)^2 + t$. Draw the *extensive form* for the game with the payoffs at each end node.

3. Let G be the two-person game in which the pure strategy sets are $S_1 = S_2 = [0, 1]$, and the utility function for each player i is $u_i(s_1, s_2) = s_1 s_2 - 0.1(s_i)^2$. Show that (0, 0) and (1, 1) are the only (pure) Nash equilibrium of G.

4. Consider a two-player game in which player 1 chooses the strategy x_1 from the closed interval [-1, 1] while player 2 chooses the strategy x_2 from the same closed interval [-1, 1]. Player 1's utility function is $x^2_1/2 - x_1x_2$ and player 2's utility function is $x^2_2/2 + x_1x_2$. Find and plot the best-response function of each player (against any pure strategy of the opponent). Is there a pure strategy Nash equilibrium of the game?